ModelIt!

Student Activity Pack

Why Baking Soda Explodes: Reaction Rates and Energy

How Reactant Amount and Surface Area Control Chemical Reactions

Grade 08L2 | NGSS MS-PS1-2, MS-PS1-5 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens when you mix baking soda and vinegar?

2 Where do the bubbles come from in a chemical reaction?

3 If a reaction produces gas, does the total mass increase, decrease, or stay the same?

4 Draw what you think happens to the atoms in baking soda and vinegar during the reaction.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Reactant Amount | Surface Area | Reaction Rate | Gas Production (CO_2) | Total System Mass

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Standard Reaction

Moderate amounts, medium surface area

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Explosion

Maximum reactant, maximum surface area

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Slow Fizz

Low reactant, low surface area

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Temperature — Higher temperatures increase molecular motion and speed up reactions

Catalyst Presence — Substances that speed up reactions without being consumed

pH Level — Acidity of the vinegar affects how vigorously the reaction proceeds

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does baking soda and vinegar explode, and where does the mass go? Use evidence from your simulation to support your answers.

1. **Why does increasing surface area speed up a reaction even though the total amount of reactant is the same?**

2. **If mass is truly conserved, why does a fizzing open container weigh less after the reaction?**

3. **How does the baking soda reaction demonstrate that atoms are rearranged, not destroyed?**

4. **Why is it important for scientists to work with sealed systems when studying conservation of mass?**

5. **How could understanding reaction rates help in real-world applications like manufacturing or cooking?**

8 STEM Challenge: Design the Ultimate Reaction Container

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YOUR ENGINEERING MISSION

Design an experiment to prove that mass is conserved in a chemical reaction, even when gas is produced, by engineering a container that captures all products.

THE SCENARIO

Your science team has been challenged to prove conservation of mass to skeptical classmates who believe that mass 'disappears' when a reaction produces gas. Design a sealed reaction system and present your evidence.

Engineering Design Thinking

1. WHAT CONTAINER DESIGN WILL SEAL THE REACTION WHILE ALLOWING YOU TO WEIGH IT?
2. HOW WILL YOU MEASURE MASS BEFORE AND AFTER WITH PRECISION?
3. WHAT SAFETY CONSIDERATIONS ARE NEEDED WHEN GAS PRESSURE BUILDS?
4. HOW WILL YOU PRESENT YOUR DATA TO CONVINCE SKEPTICS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Chemical Engineers and Materials Scientists design chemical processes and reactions for industry, from pharmaceuticals to food production to energy. They earn $75,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In your model, both the 'Maximum Explosion' and 'Slow Fizz' scenarios produced the same final amount of CO_2 gas. Which principle does this evidence demonstrate?

- **A.** Faster reactions always produce more gas than slower reactions
- **B.** The amount of product depends on the amount of reactant, not the speed of the reaction
- **C.** Surface area determines how much product is formed
- **D.** Gas production is independent of both reactant amount and surface area

2 A student weighs a sealed bag containing baking soda and vinegar before and after the reaction. The mass stays the same. Then the student opens the bag, and the mass decreases. Which explanation is supported by the model?

- **A.** The chemical reaction destroyed some atoms when the bag was opened
- **B.** CO_2 gas produced during the reaction has mass and escaped when the bag was opened, but no atoms were created or destroyed
- **C.** The plastic bag absorbed some of the mass during the reaction
- **D.** Opening the bag allowed new atoms to enter, changing the measurement

3 Your model shows that increasing Surface Area from 20% to 100% dramatically increased Reaction Rate, even though Reactant Amount stayed the same. What particle-level explanation supports this?

- **A.** Powder has more atoms than chunks of the same mass
- **B.** Powder exposes more reactant particles for contact with vinegar, increasing collision frequency and reaction speed
- **C.** Powder is chemically different from chunks because grinding changes the molecular structure
- **D.** Powder dissolves in vinegar while chunks do not react at all

4 The model shows a NEGATIVE relationship from Reaction Rate back to Reactant Amount. What does this represent in the chemical reaction?

- **A.** Faster reactions produce less product than slower reactions
- **B.** The reaction destroys atoms, reducing the total amount of matter
- **C.** As the reaction proceeds, reactants are consumed and converted into products, decreasing the remaining reactant
- **D.** Reaction rate naturally decreases over time regardless of reactant supply

5 **Based on your model, a student wants to design an experiment that produces the most intense and fastest reaction possible. Which combination of model inputs should they use, and why?**

- **A.** Low Reactant Amount and high Surface Area, because surface area is the only factor that matters
- **B.** High Reactant Amount and low Surface Area, because more reactant always means a faster reaction
- **C.** High Reactant Amount and high Surface Area, because both inputs positively affect Reaction Rate
- **D.** Low Reactant Amount and low Surface Area, to conserve materials while still getting a reaction